

assignment - 3
3 bit Gray code to seven-segment display Decoder.

	g_3	g_2	g_1	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
1	0	0	1	0	1	1	0	0	0	0
2	0	1	0	1	1	0	1	1	0	1
3	0	1	1	1	1	1	1	1	0	1
4	1	0	0	0	1	1	0	0	1	1
5	1	0	1	1	0	1	1	0	1	1
6	1	1	0	1	0	1	1	1	1	1
7	1	1	1	1	1	1	0	0	0	0

For $a = g_2 + g_3 \odot g_1$

$g_3 \backslash g_2 g_1$	00	01	11	10
0	1	0	1	1
1	0	1	1	1

$$a = g_2 + g_3 g_1 + \bar{g}_1 \bar{g}_3$$

For b

$g_3 \backslash g_2 g_1$	00	01	11	10
0	1	1	1	1
1	1	0	1	0

$$b = \bar{g}_3 + \bar{g}_2 \bar{g}_1 + g_2 g_1 = \bar{g}_3 + (\bar{g}_2 \odot g_1)$$

For c

$g_3 \backslash g_2 g_1$	00	01	11	10
0	1	1	1	0
1	1	1	1	1

$$c = g_3 + \bar{g}_2 + g_1$$

For d

$g_3 \backslash g_2 g_1$	00	01	11	10
0	1	0	1	1
1	0	1	0	1

$$d = g_3 \bar{g}_2 g_1 + g_2 \bar{g}_1 + \bar{g}_3 \bar{g}_1$$

For e

$g_3 \backslash g_2 g_1$	00	01	11	10
0	1	0	1	1
1	0	0	0	1

$$e = g_2 \bar{g}_1 + \bar{g}_2 \bar{g}_3 + \bar{g}_3 \bar{g}_1$$

For f

$g_3 \backslash g_2 g_1$	00	01	11	10
0	1	0	0	0
1	1	1	0	1

$$f = \bar{g}_2 \bar{g}_1 + \bar{g}_2 g_3 + g_3 \bar{g}_1$$

For g

$g_3 \backslash g_2 g_1$	00	01	11	10
0	0	0	1	1
1	1	1	0	1

$$g = g_3 \bar{g}_2 + \bar{g}_3 g_2 + g_2 \bar{g}_1$$

$$g = (g_3 \oplus g_2) + g_2 \bar{g}_1$$

$$a = g_2 + g_3 \odot g_1$$

$$b = \bar{g}_3 + g_2 \odot g_1$$

$$c = g_3 + \bar{g}_2 + g_1$$

$$d = g_3 \bar{g}_2 g_1 + g_2 \bar{g}_1 + \bar{g}_3 \bar{g}_1$$

$$e = g_2 \bar{g}_1 + \bar{g}_3 g_2 + \bar{g}_3 \bar{g}_1$$

$$f = \bar{g}_2 \bar{g}_1 + \bar{g}_2 g_3 + g_3 \bar{g}_1$$

$$g = g_3 \bar{g}_2 + \bar{g}_3 g_2 + g_2 \bar{g}_1$$

$$= (g_3 \oplus g_2) + g_2 \bar{g}_1$$